About this sample:

Alongside my job market paper, I include this research proposal as a second writing sample. This is a work-in-progress draft of the second chapter described in my research statement. We are hoping to be able to export results from the Maryland Longitudinal Data System and prepare a full working paper for release in the coming months. This project is coauthored with fellow JHU Economics Ph.D. candidate Su Wang and with JHU alumnus Wonsik Ko, but substantially reflects my own writing.

Can Workforce Training Soften the Scarring Effect?

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Abstract

Workforce training programs are an affordable and accessible way for workers to develop new skills and improve their labor market prospects. This may be particularly important for workers who lose their employment: job loss has been shown to have a long-lasting scarring effect on future earnings and employment, but workers who are able to build their human capital after becoming unemployed may be able to avoid some of these impacts. Existing evaluations of workforce training programs generally use administrative data, and as a result suffer from an important shortcoming of such datasets: they typically contain total earnings, which are the product of labor supply (hours) and a wage. This is a serious limitation, since the impact of a training program depends upon which margin it operates; as we demonstrate, a positive impact on earnings is not even sufficient to show that an intervention is welfare improving. Recognizing that earnings (the product of hours and wages) places bounds on the range of each component, we utilize machine learning tools and auxiliary data to estimate the joint distribution of hours and wages conditional on earnings and a sparse set of shared covariates. We demonstrate that our method performs well by presenting three out-of-sample (and out-of-dataset) validation exercises replicating previous results. Finally, we apply this method to administrative data from Maryland to study the impact of workforce training programs on the wages and hours of workers who lose their jobs in the COVID-19 recession, overcoming a limitation of previous such analyses.

Keywords: Workforce training; administrative earnings data; data combination

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1 Introduction

It is well-documented in the literature that job loss often leads to a protracted *scarring effect*: workers who suffer from unemployment go on to experience lower labor supply and wages. This is true both in the short run and in the long run (Lachowska et al. 2020; Huckfeldt 2022; Schmieder et al. 2023). This may be due to a loss of human capital (which atrophies during spells of unemployment), loss of connections to employers, or because search frictions and time-limited unemployment benefits result in unemployed workers accepting worse employment than they had prior to their job loss.

Workforce training programs may offer an affordable and accessible path for workers to avoid the worst of these impacts.[5] By developing new skills, individuals may counteract the loss of human capital from spells away from work. Career-oriented training programs with direct pipelines to employers, such as those in the building trades or skilled nursing, may provide their participants with better connections to employers which help them avoid accepting a sub-optimal position when they are ready to move back into work. Evidence on education and training after job loss is mixed. Studies of community-college schooling and post-secondary enrollment find later earnings gains among displaced or unemployed workers (Jacobson et al. 2005; Leung and Pei 2023). By contrast, Heinrich et al. (2013) find small and statistically insignificant earnings gains from participation in the WIA Dislocated Worker program, despite employment gains, whereas participation in the WIA Adult program improves both earnings and employment. Their separate comparison of trainees with participants receiving other WIA services also finds no clear additional earnings benefit from training for dislocated workers three to four years after program entry.

We use "workforce training" to refer to terminal programs, generally resulting in a diploma or a certificate, at the sub-Associate's degree level. These may range in duration from several months up to two years, and are often (but not always) provided at community colleges and other institutions of higher education. Previous work discussed above has evaluated the impacts of such programs; a related literature examines the labor-market returns to certificates and other credentials among the broader population, not restricted to displaced workers. Using Kentucky administrative data, Jepsen et al. (2014) find substantial earnings gains from diplomas but smaller returns to certificates, with considerable variation by field of study. Stevens et al. (2019) similarly find positive earnings returns to career technical education credentials in California, with the largest gains in health-related programs. Focusing on noncredit workforce training in Virginia, Xu et al. (2026) compare credential earners with individuals who complete the same training without obtaining a credential and estimate an increase in quarterly earnings conditional on employment, partly attributable to

[5] In Maryland, where we will focus our work, the state Promise scholarship makes such training effectively free for the vast majority of residents.

transitions into higher-paying industries.

We seek to contribute to this literature in several ways. First, we exploit variation in access to training following the onset of COVID-19. Many workers lost their jobs as a result of pandemic-related closures, but during the subsequent recovery workers in different geographic areas had varying levels of access to hands-on training due to location-specific restrictions on in-person gatherings (Schanzenbach and Turner 2022). This natural variation will allow us estimate the causal impact of training. Second, we use administrative data from the state of Maryland (Maryland Longitudinal Data Set, or MLDS) which contains detailed information on the type of program, number of credits attempted, and number of credits completed. This allows us to estimate program-specific treatment effects which can vary with intensity; it also allows us to compare individuals who enroll, those who partially complete, and those who successfully complete a given training sequence, going a step beyond Xu et al. (2026). Because the number of workers who participate in any specific training program is low, survey datasets would be vastly under-powered to detect positive effects. We are able to do so because our dataset contains information on the full population of Maryland workers.

Our third contribution begins with a methodological advance, but ultimately allows us to make a substantive economic contribution. Many administrative-data evaluations examine earnings and employment without separately observing hours worked and hourly wage rates (Heinrich et al. 2013; Jepsen et al. 2014; Stevens et al. 2019). This is because administrative data generally uses workforce outcomes from state Unemployment Insurance systems, which only contain information on total earnings. The value of distinguishing the treatment effect on hours versus wage is illustrated by Jacobson et al. (2005), who examine both margins using administrative data from Washington state. Evidence from sectoral employment programs further highlights this distinction: Katz et al. (2022) report that earnings gains are substantially driven by movement into higher-wage jobs, rather than increases in employment and hours alone.

An increase sheds no light on which margin the intervention operates. It may be that training increases human capital and allows workers to earn a higher hourly wage, but it also may be that graduates simply provide more hours of work. This could occur because they have better connections to the labor force, or because they must work more hours to make up for their lost earnings and other costs incurred during training. Policymakers considering an expansion of or subsidy for training need to know how much the program operates on each margin in order to find the optimal level and structure of their new policy.

To address this, we begin with the simple observation that labor earnings Y are the product of the hourly wage W and labor supply H; in other words, $Y = H \times W$. Thus, if we want to impute the values of H and W gien Y, we are already in a substantially better position than is typically the case in the data combination literature (Ridder and Moffitt 2007). We utilize an auxiliary dataset, in

this case the Current Population Survey (CPS), which contains H and W separately. We focus on a sparse set of covariates X which are shared by the CPS and the MLDS: this includes, for example, gender, race, age, and education. We estimate a labor supply function $H = h(W; X)$ which tells us the optimal choice of labor supply at any given wage, for a person of characteristics X. We can then take the estimated function to the administrative data and back out our best-guess of $\hat{H}$ and $\hat{W}$ given X by solving for $\hat{W}$ in:

$$Y = \hat{W} \times h\left(\hat{W}; X\right)$$

In other words, we ask: *Given this person's level of earnings and their observable characteristics, what do we think their hours and wage must have been?* And we answer this question based on the CPS model.

We perform three validation exercises. First, we show the fit using a hold-out sample from the CPS. We replicate the findings of Beland (2015). We then perform two true out-of-sample exercises by using our methodology to replicate prior findings in two additional datasets which, crucially, *do* contain H and W separately (and thus allow us to truly validate the accuracy of the method). We first replicate the findings of Arcidiacono et al. (2010), who leverage the NLSY79. We then replicate Huckfeldt (2022), who uses the PSID.

With the imputation methodology established, we can then estimate the impact of workforce training for displaced Maryland workers not only on Y, but also on $\hat{H}$ and $\hat{W}$, taking into account the additional uncertainty created from the first-stage imputation. We do so by leveraging access to in-person training programs as described above, and assess robustness using a combination of exact matching on selected pretreatment characteristics and propensity-score matching, following previous evaluations of training programs (Rinne et al. 2011; Biewen et al. 2014). Beyond the current application, we hope this method will be useful for other researchers using administrative data who may also benefit from examining hours and wages separately, rather than just focusing on earnings.

This project is still a work-in-progress. Ultimately, the paper will proceed as follows. In Section 2, we detail the imputation methodology and show how we apply it to the CPS. In Section 3, we present the validation exercises, establishing its reliability. We then turn to the empirical application. We discuss the MLDS in Section 4, explain our empirical strategy in Section 5, and then show the results in 6. Section 7 concludes with discussion of future directions.

2 Imputing Hours and Wages

2.1 Labor Supply

Individuals are described by a vector of characteristics X_i, which includes factors such as their age, gender, and education. Their wage offer is drawn from a distribution which depends on these observables:

$$\ln w_i = g(X_i) + \eta_i \tag{1}$$

Where $g(\cdot)$ gives the deterministic portion of wages, and η_i represents a random draw from a generic distribution where $\mathbb{E}[\eta_i] = 0$.

Agents receive a reward $u(h_i; w_i, X_i, \chi_i)$ based on their labor supply h_i, where χ_i represents unobserved heterogeneity and could reflect an individual-specific stochastic component. Individuals observe w_i and then choose labor supply h_i by solving:

$$h_i^*(w_i; \chi_i, X_i) \in \arg\max_{h_i} u(h_i; w_i, X_i, \chi_i) \tag{2}$$

This could be interpreted as a static labor supply problem, where the only control variable is labor supply. Alternatively, this is the setup that would result from a dynamic model where workers follow an exogenous decision rule for consumption. Most commonly this would represent hand-to-mouth behavior, with individuals consuming their income in each period, but it could also be that agents use some rule of thumb for what fraction of income to consume. In either case, the only decision individuals face is how much to work, with consumption and savings behavior taken as given. This is equivalent to saying that the dynamic problem of choosing an optimal h_{it} in each year reduces to a series of static decisions where the choice in period t does not depend on outcomes in future periods $t + s$.

This is a strong assumption, and since we do not restrict ourselves to a low-income sample (sometimes a justification for the hand-to-mouth assumption) this is not our preferred interpretation. However, it is not feasible for us to solve a full dynamic model of labor supply and savings; besides concerns about computational feasibility, administrative data (and most survey data!) do not contain information on assets which is normally required for identifying consumption and savings behavior.

Instead, we prefer a third interpretation, in which eq. (2) is a *nested* decision which may follow from other control variables. More generally, we may think that individuals solve the following dynamic problem:

$$\begin{aligned} \max_{h,a} V_t(h,a) &= \max_{h,a} \{u(h,a) + \mathbb{E}V_{t+1}(h,a)\} \\ (h^{**}, a^{**}) &\in \arg\max_{h,a} V(h,a) \end{aligned} \tag{3}$$

Here, individuals choose h_t as well as other control variables a_t (e.g. consumption/savings) to maximize the sum of properly discounted expected utility across the life cycle. The model of equation (2) is then nested as choosing the optimal h_t after a_t has been decided. If the included state variables X_i contain sufficient information for the choice of a_t, then our method can be interpreted as learning the policy function for h within the full dynamic problem, even though we do not model it explicitly. This is equivalent to the assumption that:

$$\begin{aligned} \arg\max_h u(h) &\subseteq \arg\max_h V(h;a) \\ &\text{so that} \\ h^* &\in \{h^{**} : (h^{**}, a^{**}) \in \arg\max_{h,a} V(h,a)\} \end{aligned} \tag{4}$$

So the joint maximization solution is the same as a procedure of nested maximization: first over other choices, and then over hours to work.[6]

The econometrician's problem

From the point of view of the decision-maker, there is no uncertainty in h_i; agents receives a wage offer w_i and make a choice h_i. This may be because they have full information (they observe χ_i) or they may be integrating out remaining uncertainty. But they know their choice. Here and in what follows we shift to the point of view of the *econometrician*, who faces significant uncertainty.

If we observe only X_i, the model suggests predicting h_i by working with the joint distribution $f(w_i, \chi_i | X_i)$. This would require estimating the wage equation (1) and placing some assumptions on η_i, specifying a utility function and solving the maximization problem to get the policy function, and placing some assumptions on the unobserved component χ_i. In this way we could derive a joint probability of w_i and χ_i (or equivalently of η_i and χ_i), and then calculate the expected value of h_i as:

$$\mathbb{E}\left[h_i(w_i;\chi_i,X_i)|X_i\right] = \iint_D h^*(w_i;\chi_i,X_i)\, dF(w_i,\chi_i|X_i) \tag{5}$$

Where D is the domain of integration for the space of χ_i and w_i values: our specification of eq. (1) implies $w \in \{0, \inf\}$, and we let χ take on full support in $\mathbb{R}$. Intuitively: we consider what the optimal h_i would have been over all the possible joint realizations of χ_i and w_i (remembering that the uncertainty in w_i comes from a shock η_i, as in eq. (1)), and take a weighted average based on the probability of each of those realizations. Luckily, our situation is not quite so dire.

[6] We do not consider the case where individuals are at the corners ($h = 0$), or deal with selection into work more broadly, since the problem we are studying only concerns people for whom $h > 0$.

2.2 Imputation with Y_i

We now consider how the econometrician can approach the case where $Y_i = h_i \times w_i$ is observed, but h_i and w_i are not seen independently. We will first consider an intuitive explanation.

Knowing Y_i bounds the values of h_i and w_i to the curve $Y_i = w_i \times h_i$; we denote this set of permissible points as $\Gamma_{Y_i} = \{(w_i, h_i) : Y_i = w_i \times h_i\}$. Acceptable values of h_i will asymptote to 0 as $w_i \to \infty$, and to ∞ as $w_i \to 0$. This is shown in figure 1.

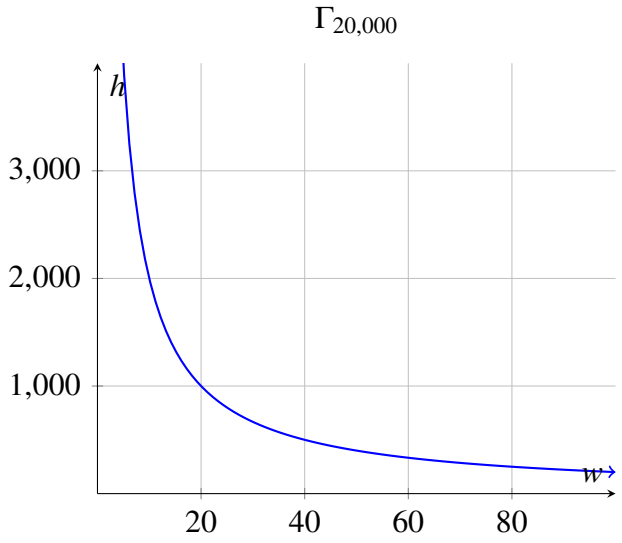


Figure 1: Possible h_i and w_i at an income of \$20,000.

We can go a bit further based on our economic knowledge. Specifically, we can constrain wages to be higher than some minimum level $\underline{w}$, and we might constrain hours to be less than some maximum level $\overline{h}$.[7] Figures 2 then shows how the picture might change somewhat; in this case, constrained by the minimum wage, though with a higher income level the maximum level of hours would be the constraining factor.

Now recall the policy function from eq. (2), which we will call $h^*(w_i; \chi_i, X_i)$. This is a mapping from w_i to h_i given X_i and χ_i. By finding the point or points where $h^*(w_i; \chi_i, X_i)$ intersects with Γ_{Y_i}, we find the h_i choice which necessarily generated the data Y_i. This can be seen in figure 3; in this case, h_i is point identified given Y_i.

For this to be feasible, we need to approximate the policy function $h^*(w_i|X_i, \chi_i)$ and observe X_i and χ_i. Alternatively, we approximate $h^*(w_i|X_i, \chi_i)$, observe X_i, and specify $u(\cdot)$ such that $h^*(w_i|X_i, \chi_i) = h^*(w_i|X_i)$, meaning we abstract away from unobservable heterogeneity. This

[7] $\underline{w}$ may reflect a state-mandated minimum wage, or some slightly lower number which captures black market work. Hours are obviously constrained at an upper bound based on the number of waking hours in a year. We will constrain hours to be less than 4,080, which would reflect someone working 80 hours a week each week of the year.

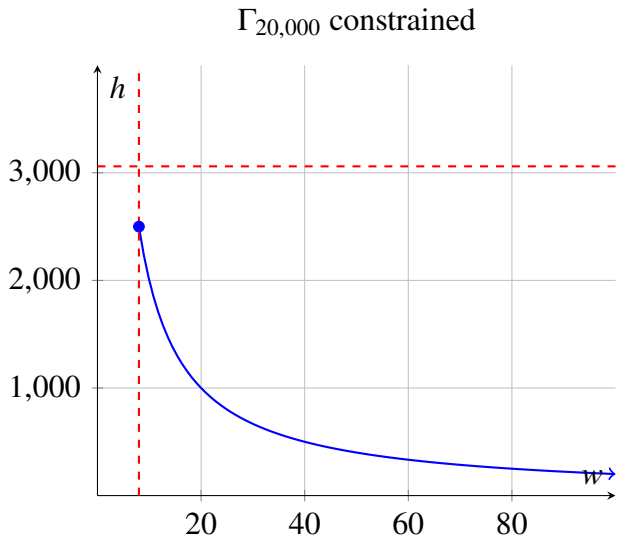


Figure 2: Possible h_i and w_i at an income of \$20,000, constrained by $\underline{w}$ and $\overline{h}$.

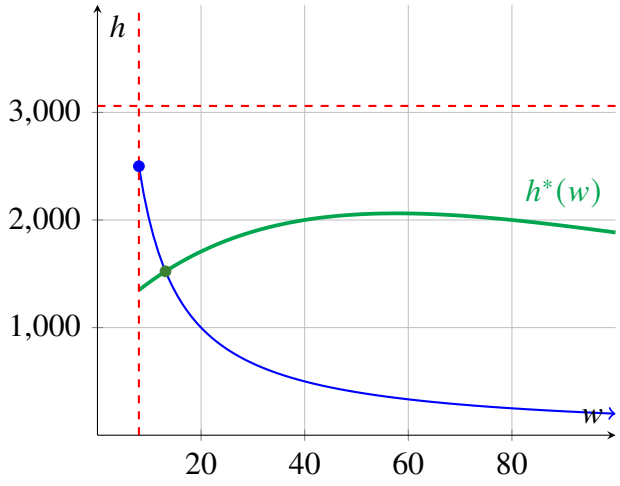


Figure 3: Point identification of h_i and w_i.

process may be problematic for various reasons; the need to parameterize and estimate h^*, to place assumptions on the error terms, and to ignore unobservables.

Without knowing χ_i, but allowing it to play a role in policy decisions, the best we can do is consider a range of possible policy functions over the unobserved variables χ_i. This comes at the cost of losing point identification, and results in the situation shown in figure 4.

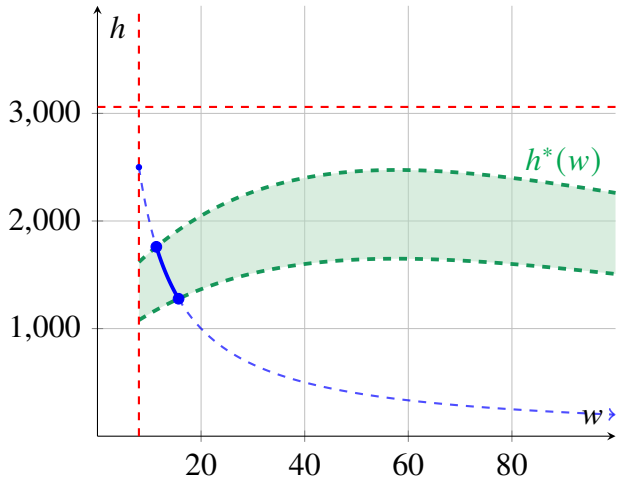


Figure 4: Identified set (solid blue region) of h_i and w_i given X_i.

We now have a set of possible (w_i, h_i) points in Γ_{Y_i}. The probability of each having been our choice depends on the joint distribution of χ_i and w_i, and by integrating over this region we can find the expected level of labor supply and wage.

We will now turn to the statistical approach. Our problem is, given Y_i and X_i, to find w_i and h_i which led to Y_i. We have shown that if there is no unobserved heterogeneity (eg the policy function is *deterministic* given w_i and X_i from the point of view of the econometrician), then under very general conditions for h_i^*, there is a unique intersection between the feasible set Γ_{Y_i} and the policy function $h_i^*(w_i; X_i)$. For example, if $h_i^*(w_i) = w_i^2$, then $Y_i = w_i \times w_i^2$, $w_i = Y_i^{1/3}$, and $h_i = h_i^*(Y_i^{1/3}) = Y_i^{2/3}$. That is to say, if there is no unobserved heterogeneity in the policy function (no uncertainty for the econometrician), then knowing Y_i tells us what η_i was, and we know both w_i and h_i.

However, we wish to proceed under the more general assumption that, for the econometrician, χ_i makes $h^*(w_i; \chi_i, X_i)$ a random variable which yields a distribution of possible h_i values for given w_i and X_i; in this case, point identification is not possible. But knowing Y_i lets us improve significantly on the unconditional expectation in eq. (5).

Analogously to Γ_{Y_i} but now using our random variable directly, define $\Omega_{Y_i} = \{(w_i, \chi_i) : w_i \cdot h^*(w_i, \chi_i, X_i) = Y_i\}$. We assume that h^* is monotonic in χ_i, so for any given (w_i, X_i, Y_i), we can solve this equation and get a unique $\chi_i = \chi_i(w_i, X_i, Y_i)$, where $\chi_i(w_i', X_i, Y_i)$ is defined as the

$\chi \in \Omega_{Y_i}$ such that $h^*(w_i', \chi_i, X_i) = Y_i$. Incorporating Y_i into eq. (5) we have:

$$\mathbb{E}\left[h_i | Y_i, X_i\right] = \iint_{\Omega_{Y_i}} h^*(w_i, \chi_i; X_i)\ dF_{w,\chi}(w_i, \chi_i | Y_i, X_i) \tag{6}$$

Luckily, we do not have to work with the joint distribution of w_i and χ_i directly. Thanks to our constraint, $Y_i = w_i \cdot h_i^*(w_i; \chi_i, X_i)$, we will instead be able to work with the marginal distribution of either w_i or χ_i. Intuitively, this is because conditional on Y_i, for any given w_i, we know what h_i must have been and thus what χ_i must have been; there is only one degree of freedom in the problem once we condition on Y_i.

We first rewrite eq. (6) using the marginal densities:

$$\begin{aligned}\mathbb{E}\left[h_i | Y_i, X_i\right] &= \iint_{\Omega_{Y_i}} h^*(w_i, \chi_i; X_i)\ f_{w,\chi}(w, \chi | Y_i, X_i)\ d\chi\ dw \\ &= \iint_{\Omega_{Y_i}} h^*(w_i, \chi_i; X_i)\ f_{\chi|\cdot}(\chi | w_i, Y_i, X_i)\ f_{w|\cdot}(w | Y_i, X_i)\ d\chi\ dw\end{aligned} \tag{7}$$

We take two further steps to simplfiy. We substitute in $\chi_i = \chi_i(w_i, Y_i, X_i)$ so that h^* is not a function of χ at all. We also wish to write out the bounds for w and χ separately. Abstracting from a minimum wage and the maximum number of waking hours in a year, w may take on any value from 0 to ∞: as $w \to 0$, $h \to \infty$ which given monotonicity of h in χ implies either that $\chi \to 0$ or $\chi \to \infty$; and then vice versa for the case where $w \to \infty$ (later imposing a maximum bound on h and a minimum wage on w will not alter the logic). However, for any given w_i, our setup implies a unique value $\chi(w_i, Y_i, X_i)$. So we can rewrite eq. (6) as:

$$\begin{aligned}&= \int_0^\infty \int_{\chi(w_i, Y_i, X_i)} h^*(w_i, \chi(w_i, Y_i, X_i); X_i)\ f_{\chi|\cdot}(\chi | w_i, Y_i, X_i)\ f_{w|\cdot}(w | Y_i, X_i)\ d\chi\ dw \\ &= \int_0^\infty \underbrace{\left(\int_{\chi(w_i, Y_i, X_i)} f_{\chi|\cdot}(\chi | w_i, Y_i, X_i)\ d\chi\right)}_{\equiv \Lambda} h^*(w_i, \chi(w_i, Y_i, X_i); X_i)\ f_{w|\cdot}(w | Y_i, X_i)\ dw\end{aligned} \tag{8}$$

We wish to focus on the bracketed term Λ. Because χ_i is determined given w_i and Y_i, this is a degenerate PDF. Focusing first on the CDF:

$$F_{\chi | w_i \le w_i', Y_i, X_i}(\chi' | w_i \le w_i^*, Y_i, X_i) = P(\chi_i \le \chi_i' | w_i^*, Y_i, X_i) \tag{9}$$

For a given w_i^* define $\chi_i^* \equiv \frac{Y_i}{w_i^*}$. If χ_i' in (9) is $\geq \chi_i^*$, then $P(\chi_i \leq \chi_i' | w_i^*, Y_i, X_i) = 1$; if χ_i' in (9) is $< \chi_i^*$, then $P(\chi_i \leq \chi_i' | w_i^*, Y_i, X_i) = 0$. So $F_{\chi|w_i \leq w_i', Y_i, X_i}(\chi' | w_i \leq w_i^*, Y_i, X_i)$ is a step (Heaviside) function which is 0 whenever $\chi' < \chi_i^*$ and 1 elsewhere. The derivative $f_{\chi|w_i \leq w_i', Y_i, X_i} = \frac{d}{d\chi} F_{\chi|w_i \leq w_i', Y_i, X_i}$ is thus a delta function, which spikes to ∞ at one point and is 0 everywhere else.

Since $f_{\chi|w_i \leq w_i', Y_i, X_i}$ has no mass outside of $\chi(w_i, Y_i, X_i)$, we can extend the bounds of integration in Λ to encompass ∞ to ∞. So long as we are working with the Lebesgue integral in eq. (8), we can then write:

$$= \int_0^\infty \underbrace{\left(\int_{-\infty}^{\infty} f_{\chi|\cdot}(\chi | w_i, Y_i, X_i) \, d\chi \right)}_{\equiv \Lambda = 1} h^*(w_i, \chi(w_i, Y_i, X_i); X_i) \; f_{w|\cdot}(w | Y_i, X_i) \, dw \tag{10}$$

So that finally we have shown that:

$$\mathbb{E}\left[h_i | Y_i, X_i\right] = \int_0^\infty h^*(w_i, \chi(w_i, Y_i, X_i); X_i) \; f_{w|\cdot}(w | Y_i, X_i) \, dw \tag{11}$$

That is, the expected value of h_i is a weighted average of the conditional probability of each w_i value and the optimal level of h_i at that wage level, subject to the constraints that $Y_i = h_i \times w_i$, $w_i \geq \underline{w}$, and $h_i \leq \overline{h}$.

2.3 Specification

Now the only step required to start estimating h_i using eq. (11) is to come up with an estimate of the conditional density, $\hat{f}_{w|\cdot}(w_i | Y_i, X_i)$. We could parameterize the wage equation and derive an expression for $f(w_i | Y_i, X_i)$ analytically, or use non-parametric methods, such as estimating an empirical distribution using cell-based averages. We could then write down a utility function, and estimate a decision rule.

In what follows, we will remain agnostic about the actual form that $u(\cdot)$, $h^*(\cdot)$, and thus $f(\cdot)$ take, and instead will define a new function:

$$\phi(Y_i, X_i) \equiv \int_0^{\overline{h}} h_i^*(w_i) \, dF_{w|\cdot}(w_i | Y_i, X_i) \tag{12}$$

That is, $\phi(Y_i, X_i)$ is a mapping from observable characteristics and observed earnings to hours which embeds the above constraints. Our approach is to approximate ϕ by training a neural net ϕ^{nn} on survey data which contains the full set of variables (w_i, h_i, X_i). We can then use ϕ^{nn} on administrative data which contain only $(Y_i = w_i \times h_i, X_i)$ to come up with an estimate for h_i.

Intuitively, the survey data gives us the *joint* distribution of X_i, w_i, and h_i; we will learn that distribution and then apply it to the admin data where we only see h_i and w_i as their product Y_i. That is, we will "unscramble" the product as best we can.

We assume the individual has the following CRRA utility function:

$$V(X_i, \chi_i, h_i; w_i) = \frac{c_i^{1-\gamma}}{1-\gamma} - \alpha(X_i, \chi_i) \cdot \frac{h_i^{1+\sigma}}{1+\sigma} \tag{13}$$

For the simplified model without dynamics, we assume that consumers are hand-to-mouth, so $c_i = h_i w_i$ – they consume all of their earnings in the current period.

The first order condition tells us the optimal (static) labor supply function h^* is

$$\frac{\partial V}{\partial h_i} = w_i(h_i w_i)^{-\gamma} - \alpha(X_i, \chi_i) h_i^{-\sigma} = 0 \tag{14}$$

$$w_i(h_i w_i)^{-\gamma} = \alpha(X_i, \chi_i) h_i^{-\sigma} \tag{15}$$

$$h_i^{\sigma-\gamma} = \alpha(X_i, \chi_i) w_i^{\gamma-1} \tag{16}$$

$$h_i^* = [\alpha(X_i, \chi_i)]^{\frac{1}{\sigma-\gamma}} w_i^{\frac{\gamma-1}{\sigma-\gamma}} \tag{17}$$

2.4 Application

Using the auxiliary (CPS) dataset with information on (w_i, h_i, X_i), we estimate γ, σ, and the function $\alpha(\cdot)$ in eq. (14). We approximate $\alpha(\cdot)$ with a flexible neural network.

Because of the invertibility of (14), we can then use the estimates to back out w and h given Y. Recognizing that $w_i = Y_i / h_i^*$:

$$\hat{h}_i^* = [\hat{\alpha}(X_i, \chi_i)]^{\frac{1}{\hat{\sigma}-\hat{\gamma}}} \left(\frac{Y_i}{\hat{h}_i^*}\right)^{\frac{\hat{\gamma}-1}{\hat{\sigma}-\hat{\gamma}}}$$

$$\hat{h}_i^* = [\hat{\alpha}(X_i, \chi_i)]^{\frac{\hat{\sigma}-\hat{\gamma}}{\hat{\sigma}-1}} Y_i^{\frac{\hat{\gamma}-1}{\hat{\sigma}-1}}$$

And then $\hat{w} = \dfrac{Y_i}{\hat{h}_i^*}$, and we have finished our imputation of the two variables.

3 Validation Exercises

Here we will ultimately present three validation exercises, which replicate previous results using our imputation method:

1. Out-of-sample, within-CPS: Beland (2015)

2. Out-of-dataset, using NLSY79: Arcidiacono et al. (2010)
3. Out-of-dataset, using PSID: Huckfeldt (2022)

4 Data

We use data from the Maryland Longitudinal Data System (MLDS). Established in 2010 and begun in 2013, the MLDS unites administrative data from across the MD state government in a single longitudinal dataset containing information on K-12 education, post-secondary education, workforce outcomes, as well as other systems such as the Department of Human Services. This makes the MLDS more expansive than similar state datasets, and state officials continue to integrate more data.

Beginning in 2020, the MLDS integrated information on non-credit workforce training programs. This is an umbrella term for a set of training programs administered at community colleges which lead to certificates but do not contribute towards two- (or four-) year degrees. The dataset contains quarterly information on each individuals' course of study, number of credits, and whether or not they received a diploma.

The inclusion of data in 2020 presents a challenge, because retrospective information is not provided. That is, we may observe a worker in 2021 for whom we do not see that they complete a workforce training program. However, we cannot distinguish between a workers who truly never participated in training and one who earned a credential in 2019, before the system began incorporating this data. For this reason, we focus our analysis on young workers who could not have earned a credential prior to 2020. Our oldest workers are those who are expected to graduate high school in the spring of 2020. Because we have data through the end of 2025, we include any individual who graduated high school in the spring of 2024 (to ensure several quarters of post-graduation observations).

5 Empirical Strategy

We will ultimately follow two empirical strategies. First, we will use access to in-person training programs based on 2020 and 2021 COVID-era restrictions to introduce exogenous variation in training participation. Second, we will use exact matching methodology which follows Rinne et al. (2011) and Biewen et al. (2014).

6 Results

Here we will present results on the impact of workforce training on earnings. We will then look at the impact on imputed hours and wages.

7 Conclusion

Here we will conclude.

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A Two-period model

JOHN WILL ADD.